Answer Key - Computer Science - Undergraduate Level Assessment 2 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** A
   Options: A) Reference counting is well suited for reclaiming cyclic structures.; B) Reference counting incurs additional space overhead for each memory cell.; C) Reference counting is an alternative to mark-and-sweep garbage collection.; D) Reference counting need not keep track of which cells point to other cells.
   Note: The statement is false because reference counting cannot effectively handle cyclic structures, as the reference counts of objects within the cycle will never reach zero, preventing their reclamation.

2. **Answer:** D
   Options: A) O(log N); B) O(N log N); C) O(log N) + 0(1); D) O((log $N)^2$)
   Note: The correct answer is O((log $N)^2$) because the recurrence relation f(2 N + 1) = f(2 N) = f(N) + log N indicates that each step of the recursion adds a logarithmic term, resulting in a cumulative effect that leads to a quadratic growth in terms of log N as N increases.

3. **Answer:** D
   Options: A) choice(seq); B) randrange ([start,] stop [,step]); C) random(); D) seed([x])
   Note: The seed() function in Python 3 initializes the random number generator with a specific integer value, allowing for reproducibility in the sequence of random numbers generated.

4. **Answer:** A
   Options: A) xyz; B) xy; C) xxzy; D) xxxxy
   Note: The string "xyz" cannot be generated by the given grammar because it lacks the necessary structure defined by the production rules, specifically the requirement for the presence of non-terminal symbols B, C, and D in any generated sentence.

5. **Answer:** B
   Options: A) 3; B) 4; C) 2; D) 8
   Note: The correct answer is 4 because the ">>" operator in Python performs a right bitwise shift, which effectively divides the number by 2 for each shift, so shifting 8 (which is 1000 in binary) one position to the right results in 4 (which is 0100 in binary).

True / False

6. **Answer:** True
   Options: A) True; B) False
   Note: As N increases, the function O($N^(1/N)$) approaches 1, indicating that its growth

rate diminishes and is slower than polynomial, exponential, and other common functions, which grow without bound.

7. **Answer:** False
Options: A) True; B) False
Note: The statement is false because the expression "a[i] == max || !(max != a[i])" simplifies to "true" for any value of a[i] that is equal to max, which does not indicate inequality with max.

8. **Answer:** False
Options: A) True; B) False
Note: The statement is false because the act of logging into a system with stolen credentials constitutes unauthorized access rather than a breach of confidentiality, which specifically involves the unauthorized disclosure of information.

9. **Answer:** True
Options: A) True; B) False
Note: An Internet Protocol (IP) address is dynamically assigned to a device by a network's DHCP server when it connects to the Internet, facilitating communication between the device and other networked devices.

10. **Answer:** True
Options: A) True; B) False
Note: The presence of several different classes of registers can actually enhance the efficiency of aggressive pipelining by allowing simultaneous access to multiple types of data, thereby reducing bottlenecks and improving overall throughput in the integer unit.

Long Form Response

11. **Answer:** def text_match(text): | return ('Found a match!') | return ('Not matched!')
Note: The provided function correctly uses regex to check for the presence of an 'a' followed by zero or more 'b's in the input string, returning an appropriate message based on whether a match is found.

12. **Answer:** def surfacearea_cube(l): | return surfacearea
Note: The answer correctly outlines the structure of a function in Python, including its definition and the intent to calculate the surface area of a cube based on the length of its sides, although it lacks the actual implementation of the surface area formula.

End of Answer Key